AUTOMATING EMPLOYEE DATA

2. REQUIREMENT ANALYSIS

2.1.DATA FLOW DIAGRAM & USER STORIES

2.1.1 USER STORY

In the **Automating Employee Data** system, the HR Administrator can add, update, and delete employee details to ensure that all employee records remain accurate and up-to-date across different organizational systems. The HR team can also generate reports such as attendance, payroll, and employee summaries automatically, which reduces manual work, minimizes errors, and saves time.

The system can integrate with **Microsoft Excel, Google Sheets, HR management software like Zoho People or BambooHR, and ServiceNow** for seamless data management and workflow automation, making it easier to handle large volumes of employee information.

Employees have the ability to view their personal details, payslips, and leave balances at any time, and they can update their contact and emergency information directly in the system.

This self-service feature ensures that the HR database always contains correct and current information, reducing the need for back-and-forth communication with HR. Using **ServiceNow portals or mobile apps**, employees can access their information in real time, submit requests, and track the status of HR services efficiently.

The IT Administrator can automatically assign or revoke access to company software like **Slack, Zoom, GitHub, Microsoft Teams**, and internal tools using **ServiceNow workflows** whenever employees join or leave.

2.1.2 DATA FLOW DIAGRAM (DFD)

EXTERNAL ENTITY:

The **external entities** in the Automating Employee Data system include the HR Administrator, Employee, and IT Administrator. The HR Administrator is responsible for adding, updating, and managing employee records, as well as generating reports and monitoring data accuracy. Employees interact with the system to view their personal details, payslips, leave balances, and to update their own information when needed. The IT Administrator manages system access by assigning or revoking permissions for company software and ensures that security protocols are followed. Together, these entities interact with the system to maintain accurate, secure, and up-to-date employee data while supporting smooth HR and IT operations.

Process 1 – Employee Data Management:

This process handles all employee-related information in a centralized system. HR administrators can add new employees, update existing records, or delete outdated details, ensuring that all employee data remains accurate and up-to-date across the organization.

Process 2 – Payroll & Attendance Integration:

This process automatically synchronizes employee attendance and payroll data. By integrating with attendance systems and payroll databases, the system ensures that salaries, leave balances, and reports are accurate, reducing manual calculations and errors.

Process 3 – Document Generation:

In this process, the system automatically generates essential documents such as payslips, offer letters, contracts, and performance reports. This eliminates manual document preparation, saves time, and ensures consistent formatting and accuracy.

Process 4 – Access Control Management:

ServiceNow workflows are used in this process to automatically

assign or revoke access to company software like Slack, Zoom, GitHub, or internal tools whenever employees join or leave. This improves security, reduces manual account management, and ensures timely access for new employees.

Process 5 – Notifications & Alerts:

This process sends notifications and alerts to HR, IT, and employees whenever critical updates occur, such as a new joiner, changes in employee data, or revoked access. It helps ensure timely action, improves communication, and reduces the risk of missed updates.

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                    Start
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                      v
        HR Admin / Employee enters data
                      |
                      v
          [Employee Data Management]
                      |
                      v
        Payroll & Attendance Integration
                      |
                      v
Document Generation (Payslip, Offer Letter, Reports)
                      |
                      v
     Access Control Management (ServiceNow)
                      |
                      v
  Notifications & Alerts to HR / IT / Employee
                      |
                      v
                    End
```